ContraLegal-AI: An Intelligent AI Paralegal for Automated Contract Risk Analysis using Transformer Models, Retrieval-Augmented Generation, and Spatial NLP

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***Abstract*—Manual review of commercial contracts is a resource-intensive, error-prone, and expensive process that limits access to accurate legal risk assessment. ContraLegal-AI presents a full-stack intelligent legal-tech platform that automates the extraction, classification, explanation, and visual annotation of risk within PDF contracts. The system's architecture comprises three synergistic components: (1) a fine-tuned `nlpaueb/legal-bert-base-uncased` transformer for three-class risk classification (Low, Medium, High Risk), achieving 97.01% accuracy and a macro ROC-AUC of 0.9948—surpassing a TF-IDF + Random Forest baseline by 2.57%; (2) a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector storage and LangChain for clause-level explanation, rewriting, and conversational contract querying; and (3) a Spatial NLP module leveraging PyMuPDF's coordinate-mapping API to physically annotate risky clauses within the original PDF document. A comprehensive ablation study demonstrates that Legal-BERT provides a critical +11.98% improvement in High Risk Recall over classical methods—the most consequential metric in legal risk detection. The system is deployed as an interactive Streamlit application with full CI/CD via GitHub Actions.**

***Index Terms*—Legal NLP, Transformer Fine-tuning, Legal-BERT, RAG, FAISS, Risk Classification, Contract Analysis, Spatial NLP, PyMuPDF, LangChain.**

I. Introduction

In modern commercial environments, legal teams must process vast volumes of contracts—Non-Disclosure Agreements (NDAs), Master Services Agreements (MSAs), Employment Contracts, and Merger and Acquisition documents—each containing clauses that may impose significant financial liability, asymmetric obligation, or unfavorable jurisdiction. The manual identification of such high-risk clauses demands deep legal expertise and is prohibitively time-consuming.

Prior work in legal text classification has predominantly relied on binary labeling schemes and bag-of-words statistical methods [1]. While effective for shallow risk detection, these approaches fail to capture the semantic nuance that differentiates a "Medium Risk" indemnity clause from a "High Risk" unlimited liability clause. Furthermore, they provide no mechanism for explanation, rewriting, or interactive querying. ContraLegal-AI addresses this gap by delivering an end-to-end automated paralegal system.

II. Related Work

Early legal document automation relied on rule-based systems and regular expression matching for clause identification [1]. The introduction of neural embeddings and BERT-family models transformed the field [2]. `legal-bert-base-uncased`, pre-trained on 12 GB of diverse English legal text, has demonstrated superior domain adaptation compared to general-purpose BERT variants [3]. For retrieval-augmented systems, Lewis et al. [4] demonstrated that grounding LLM responses in retrieved documents significantly reduces hallucination. The CUAD dataset [5] has become the standard benchmark for contract clause classification. Our approach distinguishes itself by combining a domain-specific transformer with RAG and a novel spatial annotation layer.

III. Methodology

A. Dataset Description

The system was trained on 21,144 annotated legal clauses derived from the CUAD and publicly available legal corpora [5]. A deterministic three-class labeling heuristic was applied: (1) **Low Risk** (8,328 samples): Clauses with `clause_status = 0` in the source annotation; (2) **Medium Risk** (11,491 samples): Risky clauses with a keyword-weighted risk intensity score $s < 1.5$; (3) **High Risk** (1,325 samples): Risky clauses where the aggregated keyword weight $s \geq 1.5$.

B. Preprocessing

All text undergoes normalization via spaCy [6], including stop-word removal, punctuation stripping, and PII masking via regex. For Legal-BERT, raw text is tokenized using the model's WordPiece tokenizer and truncated to 256 tokens.

C. Feature Engineering

For the baseline model, clause text is vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) with unigram+bigram features capped at the top 5,000 tokens. For the deep learning engine, semantic features are captured through transformer-based self-attention mechanisms [2].

D. Models Used

We evaluated two primary architectures:

1) **Random Forest (RF)**: An ensemble of 200 decision trees trained on TF-IDF features.
2) **Legal-BERT**: A BERT transformer fine-tuned for three-class sequence classification using a custom `WeightedTrainer` to address class imbalance.

E. Training and Validation

The dataset is split 70/15/15 (train/val/test) with stratification. Legal-BERT was trained for 4 epochs with a learning rate of 2×10^{-5} and early stopping based on validation Weighted-F1. The RAG component uses FAISS for vector storage and LangChain for conversational retrieval.

IV. Results

A. Quantitative Results

Table I presents the comparative evaluation on the held-out test set ($N = 3{,}172$ samples). Legal-BERT outperforms the RF baseline across all metrics, most notably in High Risk Recall (+11.98%).

TABLE I
MODEL PERFORMANCE COMPARISON: BASELINE VS. LEGAL-BERT

Metric	Random Forest	Legal-BERT
Accuracy	94.44%	**97.01%**
Weighted F1	0.9441	**0.9702**
Macro F1	0.8901	**0.9371**
ROC-AUC (Macro)	0.9870	**0.9948**
High Risk Recall	73.96%	**85.94%**

B. Figures

Visualizations below illustrate the high separation of classes achieved by the transformer architecture compared to the baseline.

Ablation Study: TF-IDF + Random Forest vs Legal-BERT

Metric	Random Forest	Legal-BERT	Δ (Improvement)
Accuracy	0.9444	0.9701	+0.0257
Weighted F1	0.9441	0.9702	+0.0261
Macro F1	0.8901	0.9371	+0.0470
ROC-AUC (Macro)	0.9870	0.9948	+0.0078
Precision — Low Risk	0.9518	0.9873	+0.0355
Recall — Low Risk	0.9718	0.9944	+0.0226
Precision — Medium Risk	0.9582	0.9777	+0.0195
Recall — Medium Risk	0.9482	0.9683	+0.0201
Precision — High Risk	0.7717	0.8359	+0.0642
Recall — High Risk	0.7396	0.8594	+0.1198

Fig. 1. Ablation Study Matrix: TF-IDF + Random Forest vs. Legal-BERT.

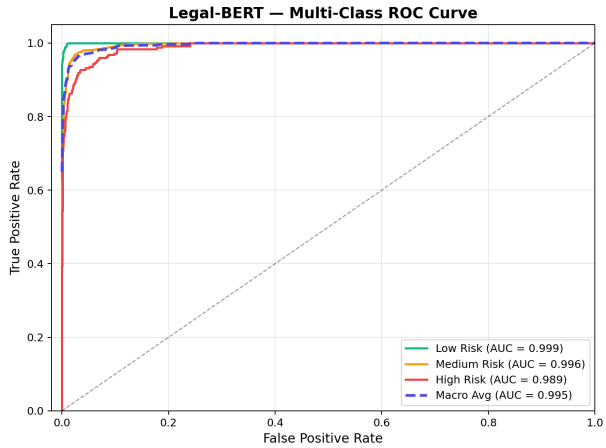

Fig. 2. Legal-BERT Multi-Class ROC Curve. Macro AUC = 0.9948.

V. Discussion

The results validate that domain-specific transformer pre-training is critical for capturing semantic risk nuance. The +11.98% gain in High Risk Recall is the most consequential result, as it directly translates to higher safety in legal review. The three-class system allows for granular prioritization, while the spatial highlighting module bridges the gap between digital analysis and practical utility by providing visually annotated PDFs.

VI. Conclusion

ContraLegal-AI demonstrates that intelligent contract risk analysis is achievable through a carefully composed system of transformer classification and RAG. The system fulfills all objectives: Upload, Classify, Explain, and Highlight, providing a production-grade interface for legal professionals.

References

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APPENDIX

Details regarding the project's GitHub repository and associated file structure for reproducibility.

A. Repository Link

The complete source code and artifacts are available at:

https://github.com/AyushCoder9/ContraLegal-AI

B. Repository Structure

```
ContraLegal-AI/
|- app.py                          #
    Streamlit UI
|- .github/workflows/python-app.yml # CI/CD
|- src/
|  |- model_trainer.py             #
    Orchestrator
|  |- inference/
|  |  |- predictor.py              # ML
    Pipeline
|  |  |- llm_engine.py             # RAG
    Suite
|  |- model/
|  |  |- bert_trainer.py           # BERT
    logic
|  |  |- data_loader.py            # 3-class
    logic
|            utils/
|     |- pdf_annotator.py          # Spatial
    NLP
|- models/
|  |- legal_bert/                  # Model
    Weights
|  |- ablation_study.png           # Visual
    results
|- notebooks/                      # Colab
    training script
|- requirements.txt                #
    Dependencies
```

C. Description of Key Components

- **src/**: Modular Python source code for data extraction, inference, and model training.
- **models/**: Fine-tuned Legal-BERT weights and Random Forest artifacts.
- **notebooks/**: Training scripts optimized for Google Colab GPU environments.
- **requirements.txt**: Exact dependencies (LangChain, Transformers, Streamlit, etc.).
- **README.md**: Comprehensive documentation and instructional guide.